

Phoebe C. Achonwa

(510) 935-5462 - pachonwa6@gmail.com - Portfolio - GitHub - LinkedIn

EDUCATION

University of California, Irvine, M. Eng in Electrical Engineering *September 2025 - December 2026*

- **Relevant Courses:** Advanced Analog Integrated Circuit Design I, Advanced Semiconductor Devices I, Micro-Sensors and Actuators (MEMS)

University of California, Santa Cruz, B.S. in Computer Engineering *September 2018 - June 2022*

- **Achievements:** 1st place in US at NeuroTechX competition - Fall
 - Contributed to the design and assembly of a non-verbal speech recognition device using facial EMG

SKILLS

Analog & Device: Analog IC design, CMOS amplifier design, differential pairs, current mirrors, biasing networks, small-signal analysis, frequency compensation, stability analysis, device modeling & scaling

MEMS: MEMS sensors & actuators, resonant structures, microfabrication concepts

EDA & CAD: Cadence Virtuoso, L-Edit, CoventorWare

Programming: Python, C (embedded), PostgreSQL, SAS, Linux

WORK EXPERIENCE

Industrial Light & Magic (Remote)

Technical Assistant I

February 2023 - September 2025

- Developed Python automation scripts to analyze and streamline workflow processes, reducing manual effort by 10% and improving productivity.
- Improved global resource management efficiency by refactoring tooling for forecasting and capturing compute and storage data, integrating data from Elasticsearch, MongoDB, and Oracle into PostgreSQL and Google Sheets using Python.

Amazon (Seattle, WA)

Software Development Engineering Intern

June 2021 - September 2021

- Built and tested backend systems in Java; gained experience in large-scale systems, code reviews, and performance analysis.

PROJECTS

Advanced Analog IC Design Project – Multi-Stage CMOS Amplifier (Irvine, CA)

- Designed and simulated a fully differential, multi-stage CMOS amplifier in 45 nm technology using Cadence Virtuoso (Spectre).
- Explored trade-offs between gain, bandwidth, power consumption, noise, and stability through topology selection, device sizing, and frequency compensation.
- Performed DC, AC, and transient simulations to evaluate amplifier performance and identify limiting factors relative to design specifications.

Mechatronics Capstone Project (Santa Cruz, CA)

Lead Developer

October 2021 - December 2021

- Designed and implemented an event-driven embedded control system using hierarchical state machines (HSMs) to coordinate sensor inputs and actuator control, interfacing analog and digital sensors with ADC-based thresholding, hysteresis, and debouncing for robust operation.
- Integrated motors, RC servo actuators, and multiple sensing modalities into a cohesive electromechanical system, debugging hardware–software interactions during bring-up and redesigning the control architecture to eliminate instability caused by excessive state complexity.